Phong Le

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EDUCATION

Duke University — Durham, NC
B.S.E. Biomedical Engineering | B.A. Computer Science — Anticipated Graduation: May 2026
Honors/Awards: QuestBridge Scholar, Florence Family Scholarship Recipient, SC TrigStar Math Competition 1st Place
Coursework: Biphotonic Instrumentation, Deep Tissue Optics, ML & Imaging, Bioinstrumentation, Signals & Systems

EXPERIENCE AND PROJECTS

No-Contact Photoacoustic Tomography with Laser Vibrometer — Durham, NC
Senior Design with Dr. Adam Wax & Dr. Junjie Yao — Jan 2026 - Present
- Using the photoacoustic effect and a laser vibrometer to non-invasively image inside tissue without requiring skin contact, enabling use on patients with fragile or compromised skin, such as neonates, burn victims, or open wounds
- Modeling full signal path from first principles (light absorption, acoustic propagation, detector response) for system design
- Integrating an optical and analog signal chain (photodiode, TIA, beam expander, pulsed laser) into an interferometer system
- Developing a Python GUI for axial image reconstruction (lateral scanning, FFT, signal averaging) targeting >1 cm depth

Photoplethysmography (PPG) Sensor — Durham, NC
Optics Bioinstrumentation Project — Aug - Dec 2023
- Built a PPG heart-rate monitor (LED/photodiode, Arduino, OLED display) with <10% error against a commercial device
- Wrote a real-time peak detection algorithm using rolling-average slope analysis to extract heart rate from noisy optical data

Hearing Aid for Low-Resource Settings — Durham, NC
Acoustic Bioinstrumentation Project — Aug - Dec 2023
- Designed and built a low-cost analog hearing aid prototype (<$30 parts), achieving <2% harmonic distortion, up to 55× gain
- Engineered a dual-passband filter system to amplify key speech frequencies while suppressing non-speech noise
- Validated performance with a 4.4/5 speech intelligibility score across user trials, in compliance with ANSI standards

Scalable Machine Learning for Antibiotic Virtual Screening — Durham, NC
Computational Drug Discovery Project — Aug - Dec 2025
- Applied transfer learning to graph neural networks, achieving ROC-AUC 0.936 on 2,560 antibiotic candidates
- Implemented robust ML validation controls (scaffold-based splits, adversarial y-shuffling) to detect data leakage

Arts North Carolina — Durham, NC
Data Analytics Project — Jan - May 2025
- Collaborated with the Arts NC director & NC DPI to produce research to support an arts education NC Bill H418
- Applied machine learning methods to identify disparities in academic performance linked to arts education
- Produced a legislative brief presented to 150+ legislators during Arts Day NC

Automated Computational Antibiotics Modeling Lab — Durham, NC
Research Project with Dr. Michael Lynch — Jan - May 2023
- Built an automated screening pipeline to find potential antibiotic candidates using molecular dynamics simulations
- Developed a Python-based workflow integrated with ChimeraX, increasing molecular simulation throughput by ~100×
- Applied AlphaFold and RDKit to perform structure-based and cheminformatics analysis on large-scale datasets

LEADERSHIP AND ACTIVITIES

MEDesign — Durham, NC
Co-President — Jan 2022 - Jan 2024
- Organized projects focused on device and digital health, coordinating mentorship & progress across teams
- Drove organizational growth from 6 to 100+ members and secured funding through multiple university partnerships
- Led a 5-person project of an app-linked wearable breathalyzer, creating a minimal viable prototype in 4 months

SKILLS

Machine Learning: Python, Scikit-Learn, RDKit, Pandas, Data Processing, NumPy, Optimization, Inverse Modeling
Research: Optics, Interferometry, Signal Processing, Data Visualization, First-Principles Modeling, Biosignal Acquisition
Embedded & Hardware: PCB Milling, Soldering, Oscilloscopes, BLE (Bluetooth), Arduino, KiCad, 3D Printing
Software & Tools: Jupyter, C, GitHub, SQL, Figma, Firebase, Flutter, Fusion 360, PyVISA, SciPy, Matplotlib, MATLAB